BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: V** **Roll No.: TYIT-11** **Date : / /2026**

Course Name: Big Data Analytics **Page no:**

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Aim :- Implement a MapReduce program that that processes a weather dataset

Steps:

- start-dfs.sh
- start-yarn.sh
- jps
- mkdir weather_TYIT-11
- cd weather_TYIT-11
- nano mapper.py

```
GNU nano 4.8                    mapper.py
#!/usr/bin/env python3

import sys
for line in sys.stdin:
    line = line.strip()
    parts = line.strip()
    if len(parts) < 2
       continue
    year = parts[0][:4]
    try :
      temp = int(parts[1])
      print(f"{year}\t{temp}")
    except valueError:
      continue
```

- nano reducer.py

```
GNU nano 4.8                    reducer.py
#!/usr/bin/env python3

import sys
current_year = None
max_temp = float('-inf')

for line in sys.stdin:
    line = line.strip()
    if not line:
       continue

    # Split the intput from the mapper by the tab character
    try:
      year, temp_str = line.split('\t')
      temp = int(temp_str)
    except valueError:
      continue
    except:
      continue

    if current_year == year:
       if temp > max_temp
```

```
GNU nano 4.8                    reducer.py
    try:
      year, temp_str = line.split('\t')
      temp = int(temp_str)
    except valueError:
      continue
    except:
      continue

    if current_year == year:
       if temp > max_temp
          max_temp = temp
    else:
       if current_year is not None:
          print("{current_yeay}\t{max_temp}")

    current_year = year
    max_temp = temp

# Output the last year
if current_year is not None:
   print(f"{current_year}\t{max_temp}")
```

- Write sample data in weather.txt file.

Teacher's Signature

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```
ubuntu@ubuntu2004:~/weather_TYIT-11$ nano reducer.py
ubuntu@ubuntu2004:~/weather_TYIT-11$ nano mapper.py
ubuntu@ubuntu2004:~/weather_TYIT-11$ echo -e "198101011200 23\n198101021200 30\r
198201011200 50\n19/weather_TYIT-11$ > weather_TYIT-35" >  weather.txt
```

- Load the data in Hadoop

```
ubuntu@ubuntu2004:~/weather_TYIT-11$ hdfs dfs -mkdir -p /user/Akshaya/weather_TYIT-11
ubuntu@ubuntu2004:~/weather_TYIT-11$ hdfs dfs -put weather.txt /user/Akshaya/weather_TYIT-11
```

- Change the mode of the file

```
ubuntu@ubuntu2004:~/weather_TYIT-11$ hdfs dfs -mkdir -p /user/Akshaya/weather_TYIT-11
ubuntu@ubuntu2004:~/weather_TYIT-11$ hdfs dfs -put weather.txt /user/Akshaya/weather_TYIT
```

- Run the cat command

```
ubuntu@ubuntu2004:~/weather_TYIT-11$ cat weather.txt |python3 mapper.py |
ubuntu@ubuntu2004:~/weather_TYIT-11$ hdfs dfs -cat | sort | python3 mapper
```

- Hadoop jar $HADOOP_HOME/share/Hadoop/tools/lib/Hadoop-streaming-3.3.6-jar -input /user/harshada/weather_TYIT03/weather.txt -output /user/harshada/weather_TYIT03/output -mapper mapper.py reducer reducer.py -file mapper.py -file reducer.py

```
ubuntu@ubuntu2004:~/weather_TYIT-11$ hdfs dfs -cat /user/Akshaya/weather_TYIT-11/out
ubuntu@ubuntu2004:~/weather_TYIT-11$ hdfs dfs -cat /user/Akshaya/weather_TYIT-11/output/part-00000
1009
```

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